# Asymmetric AlGaAs/GaAs Quantum Wells in a Dielectric Environment: Electric-Field-Tunable Optical Properties

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**Abstract.** We present a theoretical investigation of the linear and nonlinear optical properties of asymmetric AlGaAs/GaAs quantum wells embedded in a dielectric medium and subjected to an external electric field. The heterostructure, engineered through stepwise variations in Al concentration, features a four-well potential profile with intrinsic spatial asymmetry. By solving the Schrödinger equation within the envelope-function and effective-mass approximations—including self-energy corrections due to dielectric mismatch—we compute the confined electron states, dipole matrix elements, and optical absorption coefficients. Our results demonstrate that the combined influence of the dielectric environment and electric field dramatically reshapes the confinement potential, leading to Stark shifts, wavefunction relocalization, and field-dependent selection rules. The fundamental 0→1 transition remains spectrally stable in the terahertz range, while higher-order transitions exhibit pronounced blueshifts and can be selectively suppressed at critical field values where dipole matrix elements vanish. These findings highlight the dual tunability of inter-subband optical response through both electric and dielectric degrees of freedom, underscoring the potential of such structures for reconfigurable THz optoelectronic devices.

**Keywords:** Asymmetric quantum wells, Dielectric environment, Stark effect, Nonlinear optical absorption, Terahertz optoelectronics

## 1 Introduction

Scientific research into semiconductor nanostructures (SN), including quantum wells, quantum wires, rings, and quantum dots, necessitates a range of experimental, computational, simulation, theoretical, and numerical application strategies. These strategies are employed to advance the state of the art, with the objective of designing devices that are useful in a variety of cutting-edge quantum technologies [1,2]. Quantum wells (QWs) have emerged as a pivotal class of nanomaterials, exhibiting distinctive optical, electronic, and structural properties, thereby facilitating their application across a wide range of scientific and technological domains. Their tunable bandgap, high quantum yield, and size-dependent emission characteristics render them optimal candidates for optoelectronics, biosensing, and biomedical imaging [3,4,5]. The subsequent review synthesizes recent advances in QW research, grouped into five thematic areas: material synthesis and structural engineering, nonlinear optical properties, optoelectronic applications, dielectric environment effects and biomedical and biosensing applications [6,7].

Advancements in the synthesis and structural control of semiconductor nanostructures (NS) have enabled precise tuning of their properties [8,9]. In their seminal work, the researchers investigated band-edge shifts and magnetic behavior through solvation and doping strategies [10,11]. Theoretical frameworks, including the effective mass approximation and the compact density matrix formalism, were extensively employed to predict and optimize nonlinear responses, as evidenced in works by [12,13]. Employing finite element methods, the model incorporates complex geometries and diffusion effects. These studies underscore the importance of structural engineering in achieving desired optical and electronic outcomes [14,15]. Nonlinear optical phenomena, including third-harmonic generation and refractive index modulation, play a pivotal role in NS-based photonic applications. The effects of these phenomena were examined under quantum plasma and oxide confinement conditions [16,17,18]. The integration of NSs into optoelectronic devices has led to significant improvements in performance, particularly in light-emitting diodes (LEDs), lasers, and photodetectors [19]. In a similar vein, many authors investigated defect control and heavy-metal-free alternatives for QLEDs, emphasizing the pivotal role of surface chemistry and material composition in device efficiency [20,21,22].

The dielectric environment surrounding NSs exerts a pivotal influence on the modulation of their electronic and optical properties. A body of research has emerged in recent years, with studies employing theoretical models to analyze the impact of dielectric matrices and electric fields on energy levels, binding energies, and harmonic generation [23,24,25,26]. Subsequent investigations were conducted into the nonlinear susceptibility and polarizability in various core/shell configurations. These investigations revealed the sensitivity of NSs to geometric and environmental parameters[27,28,29].

Semiconductor nanoestructures have led to a paradigm shift in the fields of biomedical imaging and biosensing, owing to their unique fluorescence characteristics and biocompatibility [30,31]. A recent study investigated the theranostic applications and magnetic properties of doped NSs with the aim of utilizing them for cancer diagnostics. A number of studies have examined the integration of NSs into biosensors, emphasizing their role in enhancing sensitivity and specificity [32,33]. The following paper puts forth a series of models that predict deformation and diffusion in order to mitigate the toxicity of NS and thereby advance their clinical potential [34].

This work presents numerical calculations of the nonlinear optical absorption coefficients in asymmetric AlGaAs/GaAs quantum wells supporting at least four bound electron states, under the combined influence of an external electric field and a surrounding dielectric medium. The energy spectrum and dipole matrix elements are systematically analyzed as functions of the applied field for two distinct values of the external permittivity, representing different dielectric environments. The paper is organized as follows: Section 1 provides an introduction contextualized within recent advances in engineered quantum well systems. Section 2 outlines the theoretical model, including the effective-mass approximation, the treatment of dielectric mismatch via self-energy corrections, and the formulation of nonlinear optical absorption. Sections 3 presents the results and discussion, and Section 4 summarizes the main conclusions of the study.

## 2 Theoretical Framework

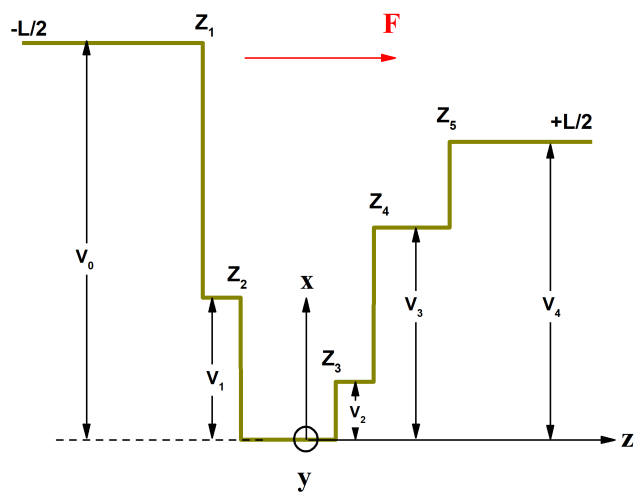


**Fig. 1.** Schematic representation of the AlGaAs/GaAs Asymmetric Quantum Well (AQW) used in this work, showing the system of reference, the direction of external electric field, with the coordinates $-L/2, Z_1, Z_2, Z_3, Z_4, Z_5, +L/2$ and the potentials $V_0, V_1, V_2, V_3, V_4$.

We consider asymmetric AlGaAs/GaAs quantum wells embedded in a dielectric medium and subjected to an external electric field. The electronic structure is calculated within the effective-mass and envelope-function approximations, assuming a parabolic conduction-band dispersion near the $\Gamma$ point. The model explicitly incorporates the effects of dielectric mismatch at the heterointerfaces, which induce image-charge–mediated self-energy corrections that reshape the confining potential.

The single-electron Hamiltonian governing the electronic states is expressed as:

$$H = \frac{p^2}{2m^*} + V(z) + e\,F\,z + \phi(z) \tag{1}$$

where $m^*$ is the effective mass of the electron in GaAs ($m^* = 0.067m_0$), assumed constant across the entire structure, $V(z)$ is the confinement potential determined by the aluminum concentration profile, $F$ is the externally applied electric field, and $\phi(z)$ is the self-energy potential arising from the dielectric environment.

The potential well $V(z)$ is defined as [35]:

$$V(z) = \begin{cases} V_0 = 282.8 \text{ meV}, & \text{if } -L/2 \leq z \leq Z_1, \\ V_1 = 101.1 \text{ meV}, & \text{if } Z_1 < z \leq Z_2, \\ V_{bottom} = 0.0 \text{ meV}, & \text{if } Z_2 < z \leq Z_3, \\ V_2 = 41.1 \text{ meV}, & \text{if } Z_3 < z \leq Z_4, \\ V_3 = 151.2 \text{ meV}, & \text{if } Z_4 < z \leq Z_5, \\ V_4 = 212.3 \text{ meV}, & \text{if } Z_5 < z \leq +L/2. \end{cases} \tag{2}$$

with $L = 60$ nm, $Z_1 = -11$ nm, $Z_2 = -7$ nm, $Z_3 = 3$ nm, $Z_4 = 7$ nm, and $Z_5 = 15$ nm.

The dielectric-induced potential $\phi(z)$ is expressed as [36]

$$\phi(z) = \frac{1}{2L_0\varepsilon_{\text{in}}} \sum_{k=0}^{\infty} \frac{(k+1)(\varepsilon_{\text{in}} - \varepsilon_{\text{out}})}{(k+1)\varepsilon_{\text{out}} + k\varepsilon_{\text{in}}} \left(\frac{z}{L_0}\right)^{2k}, \tag{3}$$

where $\varepsilon_{\text{in}}$ and $\varepsilon_{\text{out}}$ are the dielectric constants inside and outside the quantum well, respectively, and $L_0$ denotes half the distance between $Z_3$ and $Z_2$, which corresponds to the width of the central quantum well in the heterostructure [37].

The one-dimensional Schrödinger equation associated with the Hamiltonian in Eq. 1 is solved numerically using a diagonalization method. This procedure yields the eigenenergies and envelope functions $\psi_n(z)$ for various asymmetry configurations and applied electric-field strengths.

The wave function is expanded in terms of a complete set of orthonormal functions [38]:

$$\psi(z) = \sum_{n=1}^{\infty} C_n\,\varphi_n(z) \tag{4}$$

The complete orthonormal basis set is taken from the analytical solution of the infinite 1D quantum well of width $L$:

$$\varphi_n(z) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi z}{L} + \frac{n\pi}{2}\right). \tag{5}$$

Convergence of the calculated energy spectrum is achieved with $n = 50$ basis functions and a well width of $L = 60$ nm, consistent with the criteria established in previous studies [35].

### 2.1 Nonlinear Optical Absorption Model

The linear and third-order nonlinear optical absorption coefficients are calculated considering inter-subband transitions between confined states. The electric field of the incident radiation is assumed to be linearly polarized along the growth direction ($z$).

The dipole transition matrix element between initial ($i$) and final ($f$) states is defined as:

$$M_{if} = \int \psi_f^*(z)\, z\, \psi_i(z)\, dz. \tag{6}$$

The linear absorption coefficient is given by:

$$\alpha^{(1)}(\omega) = \omega e^2 \sqrt{\frac{\mu_0}{\varepsilon_0 \varepsilon_r}} \left[ \frac{\rho \hbar \Gamma_{if} |M_{if}|^2}{(E_{if} - \hbar\omega)^2 + (\hbar \Gamma_{if})^2} \right], \tag{7}$$

and the third-order nonlinear contribution by:

$$\alpha^{(3)}(\omega, I) = -\sqrt{\frac{\mu_0}{\varepsilon_0 \varepsilon_r}} \left( \frac{\omega I}{2 n_r \varepsilon_0 c} \right) \frac{4\rho \hbar \Gamma_{if} |M_{if}|^4}{\left[(E_{if} - \hbar\omega)^2 + (\hbar \Gamma_{if})^2\right]^2}. \tag{8}$$

The total absorption coefficient is expressed as:

$$\alpha(\omega) = \alpha^{(1)}(\omega) + \alpha^{(3)}(\omega, I), \tag{9}$$

where $\rho$ is the electron density, $\Gamma_{if} = 1/\tau_{if}$ is the damping factor related to the transition lifetime, and $I$ represents the optical intensity.

## 3 Results and Discussion

The combined influence of the engineered potential profile, image-charge-induced self-energy corrections, and field-induced Stark shifts leads to a highly tunable nonlinear optical absorption spectrum, as detailed in the following analysis.

**Fig.** 2 shows the confinement potential, energy eigenvalues, and corresponding probability densities for the first four bound electronic states in an AlGaAs/GaAs asymmetric quantum well (AQW), under two distinct conditions: (a) in the absence of external fields, and (b) under an applied static electric field of 40 kV/cm.

In the zero-field configuration [**Fig.** 2(a)], the ground state is strongly localized in the deepest potential well, extending from $Z_2$ to $Z_4$. The first excited state primarily occupies the adjacent well ($Z_4$–$Z_1$), while the second and third excited states are distributed across the widest region of the structure, between $Z_1$ and $Z_5$. The energy of the third excited state lies close to the top of the right-hand barrier, indicating its proximity to the continuum and weaker confinement.

Application of an external electric field of 40 kV/cm [**Fig.** 2(b)] induces a pronounced tilt in the potential profile—a hallmark of the quantum-confined Stark effect. This field-driven asymmetry effectively deepens the local well between $Z_2$ and $Z_3$, causing the ground state to shift toward this region. Likewise, the first and second excited states become more tightly confined within the $Z_4$–$Z_1$ segment. In contrast, the third excited state experiences an upward energy shift and remains localized near the upper edge of the $Z_5$–$Z_1$ well.

The potential tilt originates from a linear variation in the electrostatic potential across the structure: the energy reference at $Z_2$ is reduced to approximately $-25$ meV, whereas at $Z_3$ it rises to about $+15$ meV, with analogous shifts observed across all barrier and well regions. This linear slope leads to a modest increase in the energies of the higher-lying states—particularly the second and third excited levels. Importantly, the electron probability densities under the applied field exhibit enhanced localization within individual subwells compared to the zero-field case, reflecting the field-induced modification of the effective confinement landscape.

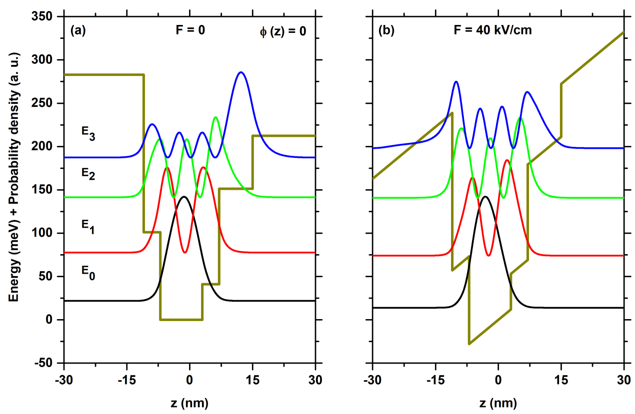


**Fig. 2.** Potential profile, energy levels, and probability density for the first four confined electronic states in an AlGaAs/GaAs AQW, for applied fields of $F = 0$ (a) and $F = 40$ kV/cm in (b), without effect of dielectric environment. The results demonstrate a high degree of congruence with those illustrated in **Figs.** 2(a, b) of the referenced study [35].

The four lowest energy levels, the non-diagonal dipole matrix elements and the total optical absorption coefficients in an AlGaAs/GaAs AQW, are presented in **Figs.** 3.

**Fig.** 3(a) illustrates the dependence of the energy levels of the first four quantum states on the applied external electric field. In contrast to a symmetric quantum well, the energy levels exhibit an asymmetric response under fields of equal magnitude but opposite polarity (e.g., at $\pm 40$ kV/cm). This asymmetry arises from the inherent asymmetry in the potential profile—specifically, the unequal energy barrier heights on either side of the well—which breaks the spatial inversion symmetry. Consequently, the ground and first excited states, whose probability densities are predominantly localized in the central and narrower region of the well, display a negative concavity in their energy-field dependence, leading to a decrease in energy with increasing field strength in one direction. In contrast, the third and fourth states, which extend more significantly into the wider regions of the potential, maintain a negative concavity, resulting in an increase in their energies under the same field conditions. This distinct behavior underscores the role of spatial asymmetry in shaping the Stark shift characteristics of confined electronic states.

The interplay between quantum well asymmetry and the applied external electric field is clearly reflected in the dipole matrix element curves shown in **Fig.** 3(b). These curves exhibit a pronounced peak for the transition from the ground state to the first excited state $M_{01}$, while transitions from the ground state to the second $M_{02}$ and third $M_{03}$ excited states yield significantly smaller matrix elements. As previously discussed, this behavior stems from the spatial localization of the electronic wavefunctions: the ground and first excited states are predominantly confined to the central region of the asymmetric well, resulting in strong overlap and hence a large dipole matrix element. In contrast, the second and third excited states extend into the lateral, wider regions of the well, where their probability densities are concentrated away from the center. This spatial separation from the ground state wavefunction reduces the overlap integral6, leading to the observed suppression of the corresponding dipole matrix elements.

**Fig.** 3(c) presents the dependence of the nonlinear optical absorption coefficients on both the incident photon energy and the strength of the applied external electric field. In accordance with the dipole matrix elements discussed earlier, the coefficients corresponding to the transitions $\alpha_{01}$, $\alpha_{02}$, and $\alpha_{03}$ are shown. All absorption peaks exhibit a blueshift as the electric field increases—an effect that is relatively modest for the $\alpha_{01}$ transition but becomes significantly more pronounced for $\alpha_{02}$ and $\alpha_{03}$. To account for the inherently small dipole matrix elements of the higher-order transitions, the vertical scale of the coefficients for the $0 \to 2$ and $0 \to 3$ transitions has been amplified by a factor of 50 relative to $\alpha_{01}$, enabling clear visualization of their spectral features. The peak positions fall within the mid-infrared region of the electromagnetic spectrum, highlighting the potential of this asymmetric quantum well system for mid-infrared optoelectronic applications.

As shown in **Figs.** 4(a, b), we examine how a dielectric environment characterized by a permittivity lower than that of the quantum well material ($\varepsilon_{\text{in}} > \varepsilon_{\text{out}}$) influences the confinement potential, the energy spectrum of the first four bound states, and the spatial profiles of their corresponding wave functions. **Fig.** 4(a), calculated at zero external electric field ($F = 0$), depicts the modified confinement potential for $\varepsilon_{\text{out}} = 3.2$, revealing that the dielectric mismatch substantially reshapes the well—enhancing its intrinsic asymmetry and altering its effective depth. When an external electric field is applied ($F = 40$ kV/cm, as in **Fig.** 4(b)), the potential profile undergoes a marked tilt, a hallmark signature of the quantum-confined Stark effect. This field-induced distortion further modulates the energy levels and wavefunction localization, underscoring the combined role of dielectric contrast and electric field in tailoring the electronic structure of asymmetric quantum wells.

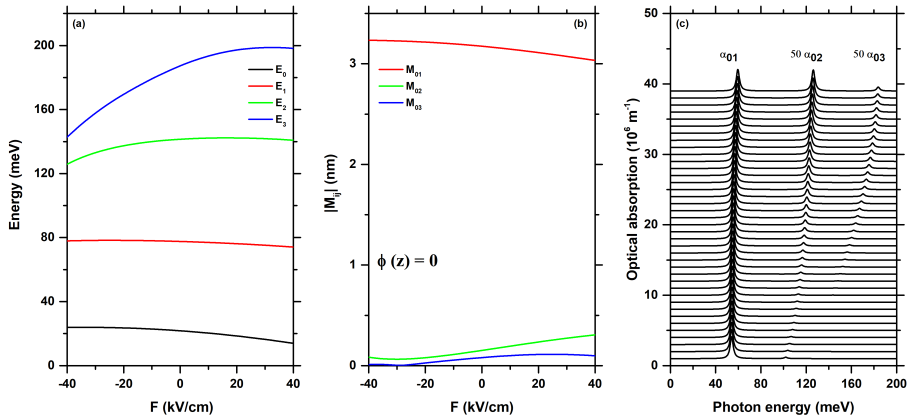


**Fig. 3.** Energy levels (a), dipole matrix elements (b) in an AlGaAs/GaAs AQW as a function of the electric field, in (c) the total optical absorption coefficients in a function of the photon energy for different values of $F$, without effect of dielectric environment.

The combined influence of the stepped square quantum well structure and the dielectric ambient potential—formulated within the framework of the image-charge model of classical electromagnetic theory—results in a modified effective confinement potential. This potential features nanoscale quasi-well regions localized between the coordinates $Z_3$–$Z_2$ and $Z_5$–$Z_4$, which significantly reshape the spatial distribution of the electron probability density for the four lowest bound states. Consequently, the envelope functions of these states deviate markedly from the characteristic sinusoidal form observed in idealized rectangular wells, instead exhibiting pronounced localization dictated by the fine structure of the engineered potential landscape. Furthermore, the effective bottom of the central quantum well is raised by approximately 35 meV relative to the original reference in **Fig.** 2(a), a shift attributed to the reduced width of the central region—now confined to roughly 9 nm—induced by the emergence of the adjacent nanonarrow wells (see **Figs.** 4) . This renormalization of the confinement potential underscores the critical role of both geometric design and dielectric environment in tailoring quantum-state properties in low-dimensional semiconductor heterostructures.

In **Fig.** 4(b), the influence of an externally applied electric field is incorporated, resulting in a pronounced tilting of the confinement potential profile—a direct manifestation of the quantum-confined Stark effect. This field-induced asymmetry lowers the energies of all four bound states and significantly redistributes the spatial probability density of the corresponding electronic wave functions. Markedly, the ground-state wave function shifts toward the left side of the central well, localizing preferentially in the region between $Z_2$ and $Z_3$. This displacement arises from the interaction between the electron's negative charge and the applied electric field, which exerts a force directed opposite to the field vector, thereby driving the electron toward the lower-potential side of the tilted well. Such field-driven spatial relocalization underscores the tunability of electronic states in asymmetric quantum well systems and highlights the intimate coupling between external fields and internal quantum confinement.

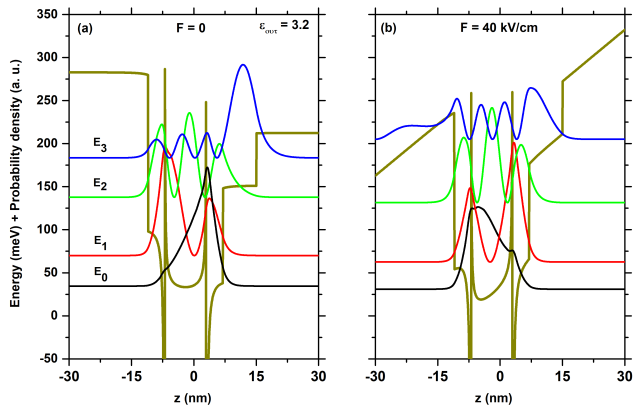


**Fig. 4.** Potential profile, energy levels, and probability density for the first four confined electronic states in an AlGaAs/GaAs AQW, for applied fields of $F = 0$ (a) and $F = 40$ kV/cm in (b), with $\varepsilon_{\text{in}} = 12.35$ and $\varepsilon_{\text{out}} = 3.2$.

The calculated energy levels and corresponding probability densities demonstrate that both the applied electric field and the dielectric contrast between the well and its surrounding medium exert a strong influence on the spatial localization of the electron wavefunctions. This modulation of wavefunction distribution directly affects the transition energies and dipole matrix elements, thereby shaping the system's optical response, show in **Figs.** 5.

Now we present in **Fig.** 5(a) the energy levels of the first four electron states confined within the quantum well, incorporating the influence of the external dielectric environment and plotted as a function of the applied electric field. Although the overall vertical energy range remains comparable to that in **Fig.** 3(a), the field-dependent behavior of the levels is markedly distinct. Markedly, the ground-state energy ($E_0$) exhibits a monotonically increasing trend as the electric field sweeps from negative to positive values, passing through a minimum near zero field. In contrast, the first excited state ($E_1$) decreases with increasing field strength but displays positive concavity, causing the energy separation $E_{01} = E_1 - E_0$ to diminish progressively with field magnitude. This reduction in interlevel spacing directly impacts the conduction-band transition energies, leading to a systematic redshift in the corresponding optical transitions. Meanwhile, the second excited state ($E_2$) shows a weak increasing trend with field, while the third excited state ($E_3$) rises more steeply than observed in the absence of dielectric contrast (cf. **Fig.** 3(a)). These modifications underscore the profound role of dielectric screening in reshaping the Stark response of confined states, thereby enabling enhanced control over transition energies and oscillator strengths for infrared and terahertz optoelectronic applications.

**Fig.** 5(b) displays the electric dipole matrix elements associated with inter-subband transitions from the ground state to the first three excited states. The matrix element $M_{01}$ exhibits the largest magnitude among the three and shows a pronounced dependence on the applied electric field: it increases monotonically with a positive slope, reaching a maximum near $F = 25$ kV/cm, beyond which it gradually decreases up to $F = 40$ kV/cm. In contrast, $M_{02}$ vanishes precisely at the same field value ($F \approx 24$ kV/cm), indicating a node in the overlap integral between the ground and second excited states under this specific field configuration. Similarly, $M_{03}$ crosses zero at $F \approx -18$ kV/cm, reflecting a field-induced symmetry condition that suppresses the dipole-allowed transition, respect a this point. These qualitative and quantitative deviations from the behavior observed in **Fig.** 3(b) arise directly from the modified confinement potential resulting from the combined influence of the stepped quantum well geometry and the dielectric contrast with the surrounding medium.

In **Fig.** 5(c) presents the nonlinear optical absorption coefficients corresponding to the inter-subband transitions from the ground state to the first three excited states. The coefficient associated with the $0 \to 1$ transition ($\alpha_{01}$) exhibits a pronounced redshift, with its resonance energy decreasing from approximately 40 meV at $F = -40$ kV/cm to about 20 meV at $F = +40$ kV/cm. This shift directly reflects the field-induced convergence of the transition energy $E_1 - E_0$, as detailed in the energy-level analysis of **Fig.** 5(a). In contrast, the $\alpha_{02}$ coefficient shows minimal variation in peak position across the field range, owing to the near-constancy of the $E_2 - E_0$ energy separation under varying electric field. Notably, the $\alpha_{02}$ peak vanishes entirely at $F \approx 25$ kV/cm—a consequence of the corresponding dipole matrix element $M_{02}$ crossing zero at this field value, as discussed in connection with **Fig.** 5(b). The $\alpha_{03}$ coefficient, on the other hand, displays a clear blueshift with increasing field, consistent with the steeper field-dependent increase of $E_3$ relative to $E_0$, leading to a growing $E_3 - E_0$ transition energy. Due to the inherently smaller dipole matrix elements for the $0 \to 2$ and $0 \to 3$ transitions, the amplitudes of $\alpha_{02}$ and $\alpha_{03}$ have been scaled up by a factor of 50 relative to $\alpha_{01}$ to facilitate clear visualization of their spectral features. Considerably, the absence of a peak in the $\alpha_{03}$ absorption spectrum at $F = -18$ kV/cm corresponds precisely to the field value at which the dipole matrix element $M_{03}$ vanishes, as shown in **Fig.** 5(b). At this specific electric field, the spatial overlap integral between the ground-state and third-excited-state wave functions becomes zero—either due to a symmetry-induced node or a field-tuned orthogonality—rendering the $0 \to 3$ optical transition dipole-forbidden. Consequently, the nonlinear absorption coefficient $\alpha_{03}$ exhibits no resonant feature at this field, underscoring the critical role of wavefunction symmetry and field-controlled localization in determining optical selection rules in engineered quantum well systems.

**Figs.** 6(a) and 6(b) illustrate the influence of a dielectric environment with $\varepsilon_{\text{in}} < \varepsilon_{\text{out}}$ on the energy levels and electron probability densities of the first four

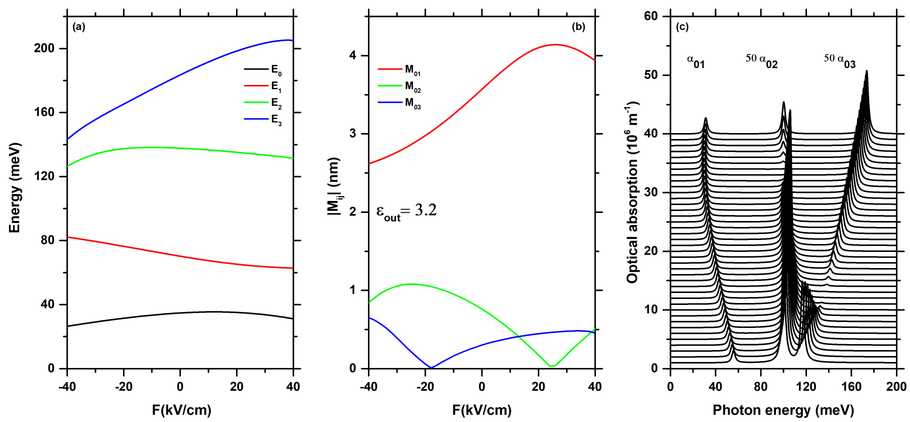


**Fig. 5.** Energy levels (a), dipole matrix elements (b) in an AlGaAs/GaAs AQW as a function of the electric field, in (c) the total optical absorption coefficients in a function of the photon energy for different values of $F$, with $\varepsilon_{\text{in}} = 12.35$ and $\varepsilon_{\text{out}} = 3.2$.

bound states in an AlGaAs/GaAs asymmetric quantum well, for zero applied electric field ($F = 0$) in panel (a) and under an external field of $F = 40$ kV/cm in panel (b).

A comparison of **Figs.** 2(a), 4(a), and 6(a) reveals the pronounced influence of the dielectric-environment-induced potential, as described by Eq. (3). In **Fig.** 2(a), this contribution is absent ($\varepsilon_{\text{in}} = \varepsilon_{\text{out}}$), yielding a flat zero-potential background. In contrast, **Figs.** 4(a) and 6(a) correspond to two limiting cases of dielectric contrast: $\varepsilon_{\text{in}}/\varepsilon_{\text{out}} \approx 4$ (i.e., $\varepsilon_{\text{in}} > \varepsilon_{\text{out}}$) and $\varepsilon_{\text{in}}/\varepsilon_{\text{out}} \approx 0.25$ (i.e., $\varepsilon_{\text{in}} < \varepsilon_{\text{out}}$), respectively. The resulting modifications to the confinement potential are striking in two respects. First, the background potential deviates from flatness: it acquires a positive concavity when $\varepsilon_{\text{in}} > \varepsilon_{\text{out}}$ and a negative concavity when $\varepsilon_{\text{in}} < \varepsilon_{\text{out}}$. Second, the potential at the center of the heterostructure ($z = 0$) shifts significantly—rising to approximately $+35$ meV in **Fig.** 4(a) and dropping to about $-10.25$ meV in **Fig.** 6(a). These changes underscore the critical role of dielectric mismatch in reshaping the effective confinement landscape through image-charge self-energy effects. An additional noteworthy feature is the effect of the dielectrically induced potential $\phi(z)$ (in Eq. 3), which effectively narrows the central region of the quantum well (This reduces the effective width to approximately 9 nm). This confinement tightening arises from the spatially varying self-energy correction associated with the dielectric mismatch at the heterointerfaces, leading to a steeper effective potential profile near the well center.

The impact of an applied electric field of 40 kV/cm can be assessed by comparing **Figs.** 2(b), 4(b), and 6(b). In the presence of dielectric contrast, the field-induced potential tilt combines with the self-energy correction to generate shallow minibands—effectively forming nanoscale quasi-wells in the $Z_2$–$Z_1$ and $Z_4$–$Z_3$ regions. This modified landscape significantly alters the spatial distribution of the electron wavefunctions. In particular, the ground state becomes localized within these newly formed quasi-wells and is further displaced toward negative $z$ due to the repulsive force exerted by the electric field (oriented along $+z$). This field- and dielectric-driven relocalization has a direct consequence on the dipole matrix elements, as will be discussed in the following paragraph.

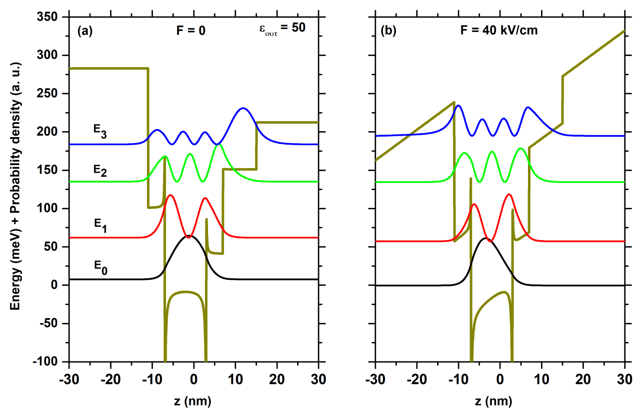


**Fig. 6.** Potential profile, energy levels, and probability density for the first four confined electronic states in an AlGaAs/GaAs AQW, for applied fields of $F = 0$ (a) and $F = 40$ kV/cm in (b), with $\varepsilon_{\text{in}} = 12.35$ and $\varepsilon_{\text{out}} = 50$.

**Figs.** 7(a)–(c) present, respectively, the four lowest energy levels, the off-diagonal dipole matrix elements, and the total optical absorption coefficients for an AlGaAs/GaAs asymmetric quantum well (AQW), as functions of the applied electric field, with a dielectric environment characterized by $\varepsilon_{\text{out}} \approx 4\,\varepsilon_{\text{in}}$.

**Fig.** 7(a) shows the electric-field dependence of the four bound states analyzed in this work. The ground state and the first excited state both decrease in energy with increasing field strength, while the second and third excited states exhibit a moderate increase—consistent with the trends observed in **Figs.** 3(a) and 5(a). The absence of symmetry with respect to the sign of the electric field reflects the intrinsic asymmetry of the initial confinement potential. Notably, all energy levels are shifted downward relative to those in **Figs.** 3(a) and 5(a), due to the combined effect of the dielectric-environment-induced potential and the applied electric field, which lower the effective reference energy of the quantum well. The weaker sensitivity of the higher-lying states to these perturbations arises because their wavefunctions are predominantly localized in the wider, less field- and dielectric-sensitive regions of the heterostructure.

**Fig.** 7(b) presents the calculated inter-subband dipole matrix elements for the case $\varepsilon_{\text{out}} > \varepsilon_{\text{in}}$, as a function of the applied electric field. As in the previously discussed configurations, the $M_{01}$ element dominates, exceeding 3 nm in magnitude over a wide field range. In contrast, the $M_{02}$ and $M_{03}$ matrix elements remain below 0.2 nm and exhibit distinct zeros at $F \approx 2$ kV/cm and

$F \approx -18$ kV/cm, respectively. These nodal points arise from the spatial separation between the ground-state wavefunction—localized in the central well—and those of the second and third excited states, which are predominantly distributed across the wider lateral regions of the heterostructure. The resulting orthogonality at specific field values suppresses the corresponding dipole-allowed transitions.

In **Fig.** 7(c) displays the nonlinear optical absorption coefficients as a function of incident photon energy, computed over the full range of applied electric fields ($-40$ to $+40$ kV/cm) for the inter-subband transitions from the ground state to the first three excited states. These coefficients are calculated using the transition energies $E_{if} = E_f - E_i$ derived from **Fig.** 7(a) and the squared dipole matrix elements $|M_{if}|^2$ shown in **Fig.** 7(b).

The resonance peak associated with the $0 \rightarrow 1$ transition ($\alpha_{01}$) remains nearly invariant across the field range, shifting only slightly from 58 meV at $F = -40$ kV/cm to 59 meV at $F = +40$ kV/cm. This weak field dependence reflects the almost constant energy separation $E_1 - E_0$. In contrast, the $\alpha_{02}$ and $\alpha_{03}$ coefficients exhibit a pronounced blueshift with increasing field, as the energies $E_2$ and $E_3$ rise with positive slopes while $E_1$ (and to a lesser extent $E_0$) decreases.

Due to the significantly smaller magnitudes of $|M_{02}|^2$ and $|M_{03}|^2$, the amplitudes of $\alpha_{02}$ and $\alpha_{03}$ are scaled by a factor of 50 relative to $\alpha_{01}$ for clear visualization. Remarkably, spectral gaps—i.e., regions devoid of absorption peaks—appear at field values where the corresponding dipole matrix elements vanish (e.g., at $F \approx 2$ kV/cm for $\alpha_{02}$ and $F \approx -18$ kV/cm for $\alpha_{03}$), consistent with the zeros in **Fig.** 7(b). These features highlight the critical role of wavefunction overlap in governing optical activity in asymmetric, dielectrically engineered quantum wells.

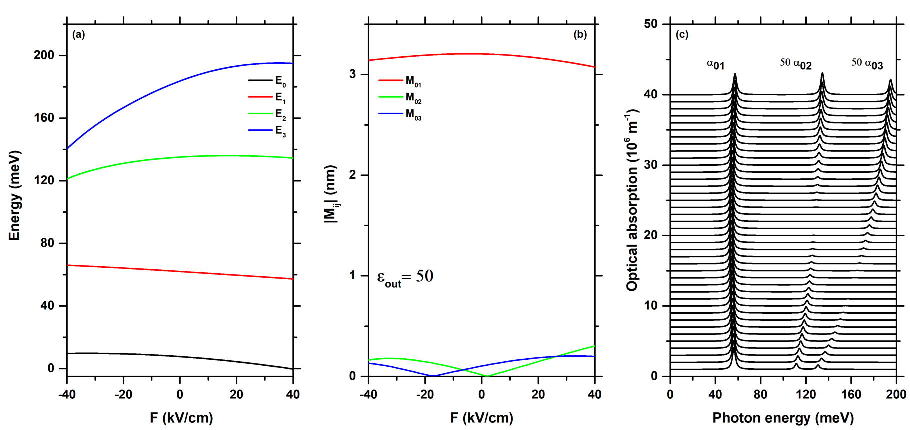


**Fig. 7.** Energy levels (a), dipole matrix elements (b) in an AlGaAs/GaAs AQW as a function of the electric field, in (c) the total optical absorption coefficients in a function of the photon energy for different values of $F$, with $\varepsilon_{\text{in}} = 12.35$ and $\varepsilon_{\text{out}} = 50$.

## 4 Conclusions

Theoretical investigations reveal that the interplay between structural asymmetry in AlGaAs/GaAs stepped square quantum wells, dielectric contrast with the surrounding medium, and an externally applied electric field offers a powerful means to engineer and dynamically control inter-subband optical transitions in the mid-infrared to terahertz (THz) range. The dielectric mismatch introduces self-energy corrections that reshape the confinement potential—creating nanoscale quasi-wells, altering the effective well depth, and shifting energy levels—while the electric field induces a tilted potential profile, triggering quantum-confined Stark shifts and field-dependent spatial relocalization of electron wavefunctions.

These combined effects lead to rich and tunable dependencies of energy spectra, dipole matrix elements, and nonlinear optical absorption coefficients on both field strength and dielectric environment. The resulting absorption spectra display well-defined, field-controllable resonances, with transition energies exhibiting either redshifts or blueshifts across the THz–infrared domain. Particularly, decreasing the external permittivity ($\varepsilon_{\text{out}}$) produces significant shifts in peak positions and modulates absorption intensities, highlighting the sensitivity of the electronic structure to dielectric screening. Moreover, at high optical intensities, nonlinear absorption becomes pronounced, and certain transitions—such as the $0 \rightarrow 2$ or $0 \rightarrow 3$ channels—can be completely suppressed at specific field values where dipole matrix elements vanish, enabling dynamic on/off switching of optical responses.

The analysis of the full field-dependent response (**Figs.** 7(a)–(c)) further reveals that while higher-energy transitions ($0 \rightarrow 2$, $0 \rightarrow 3$) exhibit pronounced blueshifts due to the stronger field sensitivity of their upper states, the fundamental $0 \rightarrow 1$ transition remains remarkably stable—its resonance energy varying by less than 1 meV over the entire field range ($-40$ to $+40$ kV/cm). This robustness, combined with the strong magnitude of its dipole matrix element, makes the $0 \rightarrow 1$ transition particularly suitable for stable THz emitters or detectors. Conversely, the field-tunable nature and selective suppression of higher-order transitions provide additional handles for designing reconfigurable optical filters and modulators. The altered potential landscape reshapes the spatial extent and effective symmetry of the wavefunctions, thereby enabling precise, field-dependent engineering of optical selection rules in nanostructured semiconductor systems.

Collectively, these findings underscore the feasibility of high-precision electro-optical modulation in asymmetric quantum well heterostructures and highlight their strong potential for next-generation, electrically and dielectrically tunable THz devices—including modulators, switches, narrowband detectors, and reconfigurable photonic components. Future efforts will extend the current model to incorporate magnetic field effects, nonparabolic band structure corrections, and direct validation against experimental spectroscopic data to further advance the design and functionality of such nanoscale optoelectronic systems.

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